Research on Cross-Spatiotemporal Remote Sensing Detection of Marine Oil Spills and Emulsions Based on Coupling Optical and Thermal Response Characteristics

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***Abstract*— Marine oil spills and emulsions are highly detrimental to the ecological environment and human health. Optical and thermal infrared remote sensing technologies provide important approaches for marine oil-spill detection, but accurate cross-spatiotemporal remote sensing detection in complex scenarios is still challenging. In this article, optical and thermal infrared observation experiments on oil spills and emulsions were designed and conducted to analyze the optical and thermal response characteristics and the technical feasibility of remote sensing detection. Then, a detection model coupling the optical and thermal response characteristics is proposed to break through the bottleneck of oil-spill detection across time and space in complex scenes, which can enhance the diversity of samples by exploiting the 3-D spatial–spectral feature (SSF) generation adversarial expansion module constrained by the sun glint intensity index (SGII). Meanwhile, based on the thermal infrared super-resolution enhancement module guided by texture features, the thermal infrared spatial geometric features of oil spills are improved, and the model's ability to resist cloud and fog interference is enhanced. Besides, under the guidance of the optical feature guidance (OFG) module, the SSF information can be extracted based on the SSF convolution deep belief network (SSF-CDBN) to realize the oil-spill remote sensing migration detection both temporally and spatially. The results show that the model can accurately detect oil spills and emulsion type, the overall accuracy (OA) and Kappa are larger than 84.46% and 0.758 in ideal scenarios, 80.11% and 0.745 in complex scenarios, which is expected to provide new technical support for oil-spill accidents.**

***Index Terms*— Deep learning, marine remote sensing, measured data, oil spills and emulsions, optical remote sensing, thermal infrared remote sensing.**

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I. Introduction

MARINE oil-spill accidents refer to maritime emergencies caused by the leakage of crude oil or other oil products in processes such as crude oil mining, processing, and transportation [1], [2], [3], [4]. Influenced by external environmental forces such as ocean winds, waves, and currents, oil spills can form various types of pollution, such as nonemulsified oil (NEO), oil-spill emulsions (OE), and thin oil films [5], [6], [7]. Alkanes and aromatic hydrocarbons in oil spills will pose risks to the life and health of marine organisms and human beings, and high-viscosity OE is more detrimental to ecological environment and is more difficult to recycle and dispose [8], [9], [10], [11].

Marine oil-spill remote sensing detection is crucial for scientific decision-making and emergency response at oil-spill accident sites, and the disposal methods for different types of OE are different. Accurately detecting the oil-spill range and OE type helps determine the scope of the accident and formulate the cleanup plan, which is also the basis for evaluating the amount of oil spilled in the accident [12], [13]. Optical and thermal infrared remote sensing technology provides an important approach for oil-spill detection. In recent years, Lu et al. [14] and Jiao et al. [15] have conducted a series of oil-spill observation experiments to investigate the optical and thermal response characteristics and mechanisms, laying a theoretical foundation for the research on oil-spill optical and thermal remote sensing detection.

Optical and thermal infrared remote sensing has technical advantages and application shortcomings. Optical remote sensing technology has been widely used in marine oil-spill detection research owing to its advantages of high spatial, hyperspectral, temporal resolution, and spatial–spectral (SS) integration [16], [17], [18]. However, it is seriously affected by sun glints and clouds, and there are obvious differences in the reflectance of targets under different sun glints [19]. Sun glints will change the positive and negative contrast of the reflectance of NEO and seawater, and will also change the spectral response of OE, bringing challenges to marine oil-spill detection. For example, in strong sun glint scenes, the reflectance of OE and seawater is positively contrasted, that is, the reflectance of NEO is higher than that of seawater,

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and it appears as a bright target in the optical image [20], [21]. Clouds and fog are also usually shown as bright targets in optical images. Based on the migration detection model without prior samples, using single-dimensional optical data to carry out oil-spill detection may cause misclassification. Zhu et al. [20] used the critical angle to characterize sun glint intensity and realized oil-spill detection under different sun glints [12]. Meanwhile, some scholars have found that texture features based on oil spills can be exploited to distinguish the oil-water boundary and describe the spatial geometric details of oil spills [22], [23]. However, limited by issues such as the spatiotemporal heterogeneity of optical remote sensing data, the bottleneck of oil-spill cross-spatiotemporal detection in complex scenarios such as the coexistence of multiple emulsion types of oil spills, cloud, and fog interference has not been effectively overcome.

Benefiting from the different radiation transmission mechanisms from optical remote sensing, thermal infrared remote sensing technology can avoid the impact of sun glints and generate an obvious response to interference targets such as clouds and fog, and it has all-day detection capabilities [24], [25], [26], [27]. Due to the high altitude and low temperature, clouds and fog appear as obvious dark targets in thermal infrared data, which makes it possible to suppress the interference of clouds and fog. However, due to the limitations of working principles, atmospheric interference, and cost-effectiveness, the spatial resolution of thermal infrared sensors is often lower than that of optical sensors, which brings challenges to eliminating the influence of clouds based on origin thermal infrared data and then accurately detecting marine oil spills [15], [28], [29]. In addition, at present, oil-spill detection research based on optical and thermal remote sensing is mostly conducted using a single technology, and the complementary advantages of optical and thermal infrared remote sensing technologies have not been fully exploited [30], [31].

Deep learning has been applied to oil-spill detection because of its powerful capabilities for model cross-spatiotemporal migration, sample expansion, super-resolution enhancement, and high-order feature extraction, thereby solving the problem of poor applicability of traditional methods in complex oil-spill scenarios [21], [32], [33]. Generative adversarial network (GAN) has been used in remote sensing training sample expansion research in recent years to address issues such as insufficient real sample data, thereby enhancing sample diversity and model generalization [34], [35]. Super-resolution deep learning models have been applied to remote sensing images recently, achieving ideal data super-resolution enhancement effects [36], [37].

This article attempts to combine the oil-spill optical and thermal response characteristics with deep learning technology to propose a cross-spatiotemporal remote sensing detection model. This model can exploit the complementary advantages of optical and thermal infrared remote sensing technologies, thereby achieving oil-spill cross-spatiotemporal migration detection in complex scenarios at different times and spaces, providing a method and data support for marine oil-spill accidents. The main objectives of this article are as follows.

1) Design and conduct oil-spill observation experiments to clarify the optical and thermal response characteristics. Calculate the sun glint intensity index (SGII), construct the oil-spill mixing spectral database (OMSD), and form the optical feature guidance (OFG) module to guide the model to detect oil-spill mixing pixels under different sun glints.
2) Take the SGII to constrain the 3-D conditional generative adversarial net (3DCGAN) to expand oil-spill samples, increase sample diversity, and enhance model generalization and detection accuracy.
3) Use texture features to guide the thermal infrared super-resolution module [TI-wide activation deep residual network (WDSR)] to improve the thermal infrared brightness temperature (BT) resolution and spatial geometric features of oil spills. Based on the oil-spill thermal infrared response characteristics to improve the model's ability to resist cloud and fog interference.
4) Use the feature extraction module based on the SS-feature (SSF) convolution deep belief network (SSF-CDBN) to couple the oil-spill optical and thermal response characteristics to realize cross-spatiotemporal remote sensing detection of marine oil spills and emulsions in different marine complex scenarios.

II. Data and Analysis

A. Acquisition of Measured Optical and Thermal Infrared Data of Oil Spills

The experiment was conducted from May 21 to May 22, 2022, at the First Institute of Oceanography, Ministry of Natural Resources, Qingdao, Shandong, China. During the experiment, the weather was sunny, the temperature ranged from 17.7 °C to 30.7 °C, the average humidity was 56.5%, the average wind speed was 3.3 m/s, the mean air pressure was 1004.6 hPa, and the average daytime light intensity was 560.6 W/m^2. The oil used is the crude oil produced by the Dongying Shengli Oilfield. To avoid emulsification of the crude oil before the experiment, dehydration treatment was performed on the crude oil used in the experiment. The asphalt content of the experimental oil is 1.35%, the density is 0.92 g/cm^3 (25 °C), the water content is 0.64%, the wax content is 11.07%, and the gum content is 28.76%.

Due to the difference in surface tension, oil and water are not easily miscible. To improve the surface tension between oil and water, referring to the research experiments of Lu et al. [14], [38], this study added emulsifiers (Tween 80 and Span 80) during the OE preparation experiment to establish a uniform and stable dispersion system. Based on the FS-1100D high-speed disperser, water-in-oil (WO) emulsion oil and oil-in-water (OW) emulsion oil with different volume concentrations were prepared by changing the volume ratio of crude oil and water, as shown in the following equation:

$$V_E = V_O/(V_O + V_W) \tag{1}$$

where V_E represents the volume concentration of OE, V_O represents the volume of crude oil, and V_W represents the volume of water.

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TABLE I
PARAMETERS OF SPECTRORADIOMETER AND THERMAL INFRARED RADIOMETER

Parameters		Index
ASD-FS4 Spectroradiometer	Spectral range (nm)	350-2,500
	Spectral resolution (nm)	3@350-1,000, 7@1,001-2,500
	Spectral sampling interval (nm)	1.4@350-1,000, 1.1@1,001-2,500
	Field angle (°)	25
	Wavelength accuracy (nm)	0.5
	Wavelength repeatability (nm)	0.1
Thermal infrared Radiometer	Resolution (pixel)	640×480
	Wavelength range (um)	8~14
	NETD (mk)	50
	Measurement accuracy (°C)	±2
	HFOV (°)	57
	Imaging method	Frame imaging

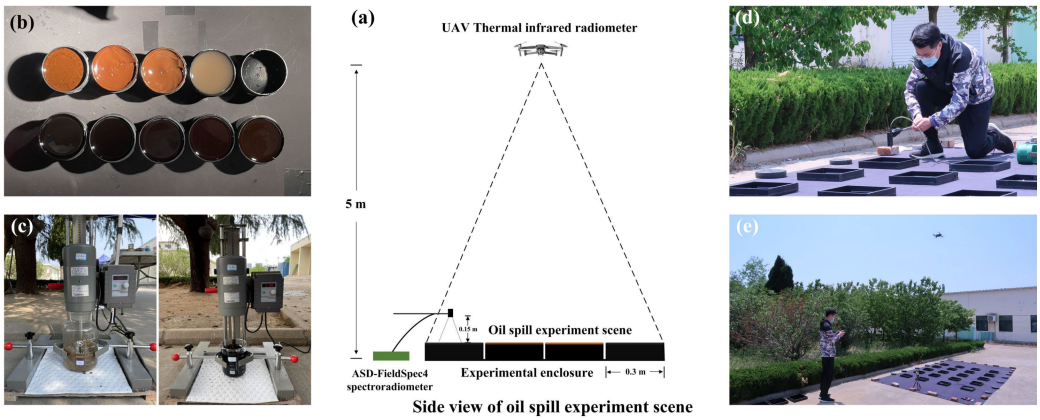


Fig. 1. Experimental schematic and photos. (a) Schematic of the experiment. (b)–(e) Photographs of the on-site experiment.

In this experiment, NEO, WO, OW, and seawater were poured into the experimental enclosure in sequence. After the oil film spread evenly and reached thermal equilibrium, experiments on optical and thermal infrared data acquisition of oil spills and emulsions were conducted by using the ASD-FS4 spectroradiometer and the thermal infrared radiometer mounted on the DJ-Mavic2 drone, and the sensor parameters are listed in Table I. In this article, the processing method of oil-spill hyperspectral data was adopted by Wang et al. [25] and [35]. The oil-spill thermal infrared data acquisition experiment lasted for 24 h, from 9:30 on the day of the experiment to 9:30 on the morning of the next day. The data acquisition interval was 0.5 h, and a total of 48 scenes of oil-spill BT data were acquired and the DJI Thermal software was used to process thermal infrared data. The experimental diagram is illustrated in Fig. 1.

B. Analysis of Optical and Thermal Response Characteristics of Measured Oil-Spill Data

Crude oil is a mixture of various liquid hydrocarbons such as alkanes, cycloalkanes, and aromatic hydrocarbons, the main elements are carbon and hydrogen, accounting for 83%–87% and 11%–14%, respectively, and the main composition is the "-CH" group. Due to the stretching vibration of groups such as "-CH," the frequency doubling absorption band is located in the near-infrared (NIR) and short-wave infrared range (780–1800 nm), and the combination frequency absorption band is located in the short-wave infrared range (1800–2500 nm). Therefore, the absorption spectral characteristics of the "-CH" group are key information for detecting marine oil spills [39].

Crude oil has the characteristics of strong absorption and low reflectance, with a strong absorption capacity of sunlight, and the incident light is difficult to penetrate the oil film and then scatter back to the sensor, leading to a weak backscattering signal of crude oil [21]. As shown in Fig. 2, the overall remote sensing reflectance of crude oil is low in the visible light range of 400–700 nm. In the near-infrared and short-wave infrared range of 700–2500 nm, the absorption characteristics of the "-CH" group are difficult to obtain due to the strong absorption. The spectral curve of seawater belongs to a typical type of case-1 water. Due to the strong absorption of water, the remote sensing reflectance of seawater in the visible, near-infrared, and short-wave infrared bands is low, and the overall reflectance is slightly higher than that of crude oil.

In this article, WO with a volume concentration of 80% and OW with a volume concentration of 0.1% were selected to perform spectral response characteristic analysis. The continuous crude oil in WO is an optically dense medium with strong light absorption ability, and the small water droplets dispersed in the oil are an optically sparse medium with a relatively low refractive index. When incident light passes from an

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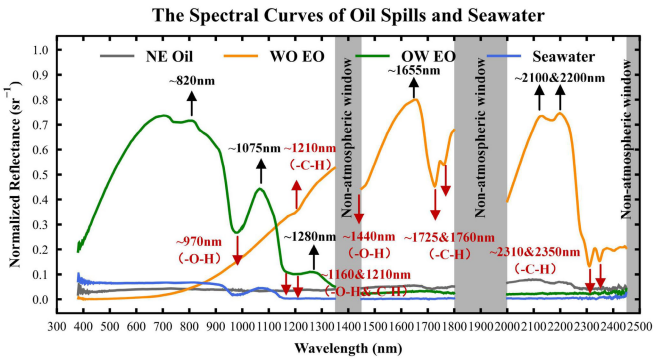


Fig. 2. Spectral curves of oil spills and seawater (the gray bands represent nonatmospheric windows).

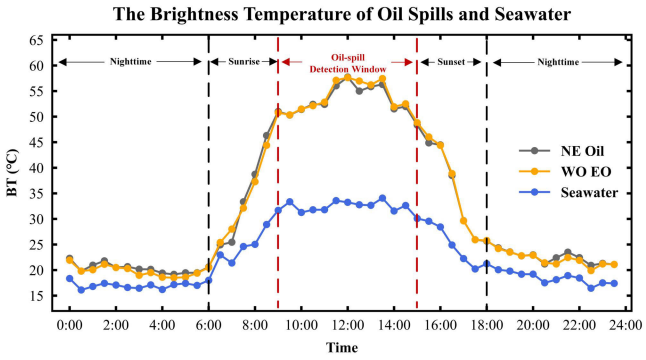


Fig. 3. BT of oil spills and seawater (time interval is 0.5 h).

optically dense medium into an optically sparse medium, small water droplets in oil will demonstrate strong scattering characteristics [14]. As illustrated in Fig. 2, the response range of WO is mainly located in the short-wave infrared band of 1300–2500 nm. There are three reflection peaks at 1655, 2100, and 2200 nm, representing the backscattering signal of small water droplets in oil. A total of five absorption valleys at 1210 nm (level-2 frequency doubling), 1725 nm (level-1 frequency doubling), 1760 nm (level-1 frequency doubling), 2310 nm (combination frequency), and 2350 nm (combination frequency) are related to the strong absorption characteristics of the "-CH" group, and the absorption valley at 1440 nm (level-1 frequency doubling) is related to the "-OH" group, which agrees with the research conclusion of Lu et al. [39].

OW is small droplets of crude oil dispersed in the continuous phase of water, and it visually resembles turbid water with high suspended solids. As shown in Fig. 2, the spectral response range of OW is mainly located in the visible and near-infrared bands of less than 1300 nm. OW has three reflection peaks at 820, 1075, and 1280 nm, representing the backscattering signals of small oil droplets in water. The troughs at 970 nm (level-2 frequency doubling), 1160 nm (level-2 frequency doubling), and 1210 nm (level-2 frequency doubling) are related to the strong absorption characteristics of the "-OH" and "-CH" groups [40]. The remote sensing reflectance of OW in the visible and near-infrared bands is higher than that of WO, and the reflectance in the short-wave infrared band is lower than that of WO.

Thus, the measured remote sensing reflectance data of NEO, WO, OW, and seawater obtained under controllable conditions have obvious differences in spectral response characteristics, which provides a theoretical basis for detecting oil spills and emulsions based on optical remote sensing technology.

The emissivity and specific heat capacity of oil spills are different from that of seawater, leading to the difference between the radiative transmission of oil spills and the atmosphere. This affects the thermal contrast characteristics of oil spills and the background seawater, thereby providing a basis for oil-spill detection based on thermal infrared remote sensing technology. Due to errors in experimental layout, this study failed to obtain effective thermal infrared BT data of measured OW. So, the thermal infrared response characteristics of NEO, WO, and seawater were analyzed to clarify the thermal infrared response mechanism and detection time window.

As illustrated in Fig. 3, since NEO and WO have similar states, the heat exchange process has a similar impact on the BT changes, and both are higher than the BT of seawater. In the daytime period (6:00–18:00), the thermal infrared response of oil spills is obvious, and the oil film has a strong absorption ability of solar radiation. After reaching the thermal equilibrium, the thermal radiation characteristics are obvious, and the BT is high. In the early morning and night periods (0:00–6:00, 18:00–24:00), due to the weak or lack of solar radiation, the thermal infrared response of oil spills is weak. The time window for oil-spill thermal infrared detection is determined to be 9:00–15:00 during the day. In this time window, there is an obvious thermal infrared response difference between oil spills and seawater.

Therefore, under controllable experimental conditions, the measured thermal infrared differences between NEO, OE, and seawater are obvious, which provides theoretical support for oil-spill detection based on thermal infrared remote sensing technology.

C. Analysis of Optical and Thermal Response Characteristics of Spaceborne Oil-Spill Data

The optical and thermal infrared sensors Thematic Mapper (TM), Operational Land Imager (OLI), and Thermal Infrared Sensor (TIRS) mounted on the Landsat-5/8/9 satellite were used to obtain optical and thermal remote sensing images of accidents such as the 2010 U.S. Gulf of Mexico oil spill, the 2017 Persian Gulf oil spill, the 2021 China Yellow Sea oil spill, and the 2022 Persian Gulf oil spill.

Landsat satellite sensors cover visible light, near-infrared, short-wave infrared, and thermal infrared bands, in which the spatial resolution of the optical image is 30 m, the spatial resolution of the thermal infrared image is 100/120 m, the satellite width is 185 km, and the revisit period is 16 days. In this article, based on ENVI software, the Landsat data was processed, and the oil-spill reflectance and BT images were obtained.

The classic scene in the "Deepwater Horizon (DWH)" oil-spill accident of the U.S. Gulf of Mexico was selected as an example to analyze spaceborne marine oil-spill optical and

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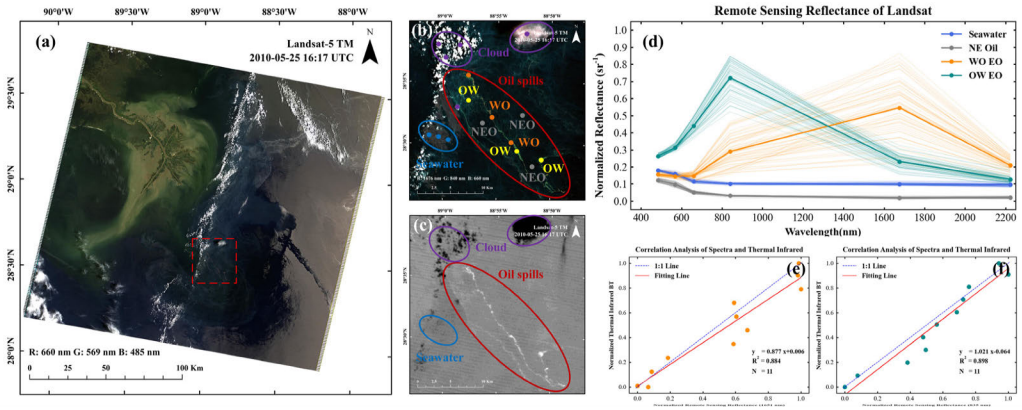


Fig. 4. (a)–(c) Oil-spill optical and thermal remote sensing images. (d) Oil-spill remote sensing reflectance of Landsat. (e) and (f) Optical and thermal response characteristics analysis results.

thermal remote sensing response characteristics, as shown in Fig. 4(a). The optical remote sensing images of the research area are represented by false colors (R: 1676 nm, G: 840 nm, B: 660 nm). As illustrated in Fig. 4(b), there are obvious visual differences between WO and OW, showing reddish brown and green colors, respectively [14], [38].

In this article, based on the oil-spill detection results of Lu et al. [14] and Jiao et al. [41], combined with our expert experience in on-site disposal of marine oil spills, the samples in the research area were manually calibrated [part of the samples were labeled on Fig. 4(b)], and samples were randomly selected for response analysis. As illustrated in Fig. 4(d), the spaceborne spectral response characteristics of WO and OW are similar to those of measured oil spills, and there is an obvious difference in spectral separability between the two. The response intervals are mainly located in the short-wave infrared and near-infrared bands, with reflection peaks appearing at 1676 and 840 nm, respectively, which is consistent with the research results of Lu et al. [14] and [15].

It can be seen from Fig. 4(c) that, in the thermal equilibrium state [15], the BT of NEO and OE is higher than that of seawater, which agrees with the measured thermal infrared response characteristics of oil spills. The BT of WO and OW in the research area is significantly higher than that of NEO and background seawater. It is believed that the thermal infrared response characteristics of OE are determined by both thickness and concentration. Due to the large amount of oil spills in this area and the high equivalent oil film thickness, there is an obvious thermal infrared radiation phenomenon in the image [15], [29].

In this article, the spectral characteristic bands of 1676 and 840 nm of WO and OW were used with thermal infrared BT, respectively, to perform correlation analysis of the optical and thermal characteristics of oil spills. As shown in Fig. 4(e) and (f), the spectral characteristic bands of WO and OW have a strong correlation with BT, and R^2 reaches 0.884 and 0.898, respectively. Meanwhile, it can be seen from this figure that cloud targets appear as obvious dark targets in thermal infrared images due to the high altitude and low temperature. In addition, because of the radiation transmission mechanism of thermal infrared remote sensing, it can overcome the influence of sun glints that are common in optical remote sensing. Therefore, it is technically feasible to jointly conduct oil-spill spaceborne remote sensing detection based on the complementary advantages of optical and thermal response characteristics.

D. Calculation of the SGII and Construction of the OMSD

Oil spills can change the roughness of the sea surface, resulting in differences in sun glint intensity, which is an important factor that changes the positive and negative contrast between the reflectance of the oil spills and the background seawater [39]. Before investigating the detection of marine oil spills, it is necessary to evaluate sun glint intensity. In this study, remote sensing observation geometric information (sun zenith angle, sun azimuth angle, satellite zenith angle, and satellite azimuth angle), surface roughness, wave inclination angle, and refractive index were used to calculate the sun glint reflectance based on the Cox–Munk model [42], as shown in (2)–(4)

$$\tan^2\beta = \frac{\sin^2\theta_0 + \sin^2\theta + 2\sin\theta_0\sin\theta\cos\phi}{(\cos\theta_0 + \cos\theta)^2} \tag{2}$$

$$P(\theta_0, \theta, \phi, \sigma 2) = \frac{1}{\pi\sigma^2}\exp\left(\frac{-\tan^2\beta}{\sigma^2}\right) \tag{3}$$

$$L_{\mathrm{GN}}(\theta_0, \theta, \phi, \sigma^2) = \frac{\rho(\omega)}{4} P(\theta_0, \theta, \phi, \sigma^2) \frac{\left(1+\tan^2\beta\right)^2}{\cos\theta} \tag{4}$$

where θ_0 represents the sun zenith angle, θ represents the satellite zenith angle, ϕ represents the relative azimuth angle, β represents the ocean wave surface inclination angle, σ represents the surface roughness of the oil spills or seawater, $\rho(\omega)$ is the Fresnel reflection coefficient (calculated from the refractive index), and $P(\theta_0, \theta, \phi, \sigma^2)$ is the probability density function calculated by the Cox–Munk model.

To reduce the complexity of judging sun glint intensity, the angle (θ_m) between the direction of sunlight reflected by the sea surface mirror and the direction detected by the satellites was taken to replace L_{GN} as the intensity index of sun glint reflectance in optical satellite images [20]. Under different sun glint intensities, the positive and negative contrast characteristics of oil film and seawater are different. In the

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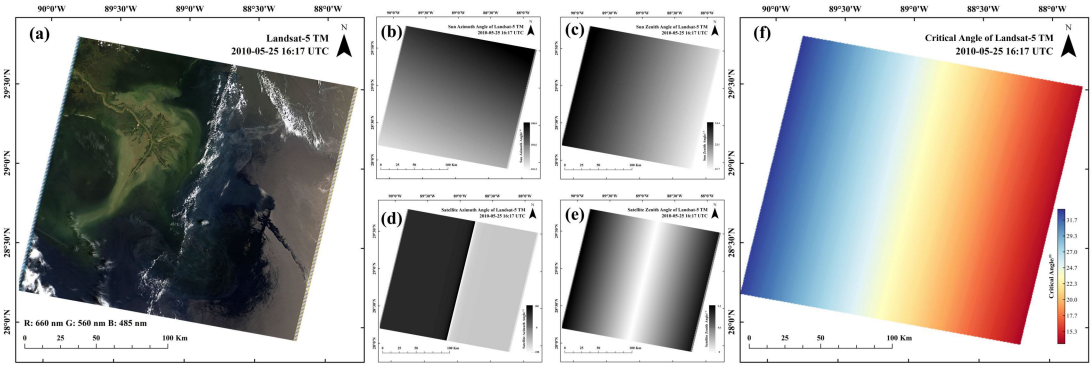


Fig. 5. Calculation of the SGII. (a) Oil-spill remote sensing image. (b)–(e) Observed geometry information. (f) SGII.

condition of a strong sun glint ($\theta_m < 12°$), the information obtained by satellites is mainly the sun glint signal, and the backscattering signal of OE (i.e., the absorption of sunlight by oil spills) is negligible. Therefore, the remote sensing reflectance of all types of oil spills is higher than that of seawater, and the oil film is in positive contrast with seawater. In the condition of a weak sun glint ($\theta_m > 13°$), the oil spills have a strong absorption effect on the sunlight, and the oil film and seawater show negative contrast characteristics [12]

$$\mathrm{SGII} = \arccos[\cos(\theta_0)\cos(\theta) - \sin(\theta_0)\sin(\theta)\cos(\phi)]. \quad (5)$$

However, the setting of the critical angle threshold is oriented to large-scale satellite images such as moderate resolution imaging spectroradiometer (MODIS) and Haiyang (HY)-1C/D, and it is not suitable for medium-scale resolution satellites such as Landsat [21]. Therefore, this article does not subjectively set a fixed sun glint intensity threshold condition for the oil-spill detection model. After the SGII is calculated, as shown in (5), it is directly used as the constraint to constrain 3DCGAN to expand the SSF cube samples under different sun glint intensities. Based on the SGII, the SSF-CDBN will be guided to achieve accurate oil-spill detection under complex sun glint conditions. The true color oil-spill image, observed geometry information, and the SGII of the research area are presented in Fig. 5.

In real marine oil-spill accidents, oil spills on the sea surface often show spatial heterogeneity and nonuniformity [38], [41]. In remote sensing images obtained by medium-resolution sensors (such as Landsat), it is difficult to have pure pixels of the same type and uniform distribution. WO and OW are mainly distinguished by volume concentration, so the mixing pixel effect will be highly obvious at the junction of OE with different emulsion types.

This article simulates the measured oil-spill hyperspectral data obtained under controlled conditions by the Landsat optical sensor based on the spectral response function (SRF) [38]. The spectral simulation formula is shown in the following equation:

$$R_n = \frac{\int_{\lambda_1}^{\lambda_2} \mathrm{SRF}(\lambda) R(\lambda) d\lambda}{\int_{\lambda_1}^{\lambda_2} \mathrm{SRF}(\lambda) d\lambda} \quad (6)$$

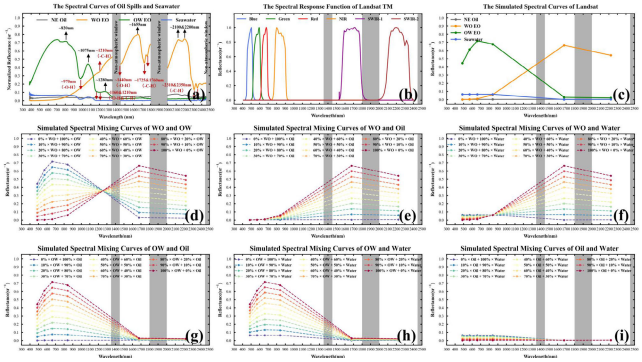


Fig. 6. Construction of the OMSD. (a) Measured oil-spill curves. (b) SRF of Landsat-5 TM. (c) Oil-spill simulated spectral curves. (d)–(i) Simulated spectral mixing curves.

where R_n represents the simulated reflectance of the nth band, $R(\lambda)$ represents the measured oil-spill hyperspectral reflectance, $\mathrm{SRF}(\lambda)$ is the spectral response function, and λ_1 and λ_2 represent the wavelength range of each band.

As shown in the following equation, based on the linear spectral mixing model, OMSD is constructed according to the horizontal area fraction k of each oil-spill target in the mixing pixel, and the spectral response characteristics of oil spills are employed to guide the detection model to detect oil-spill mixing pixels in real oil-spill scenarios:

$$R_{\mathrm{Mix}} = k_1 \cdot R_{\mathrm{Oil}} + k_2 \cdot R_{\mathrm{WO}} + k_3 \cdot R_{\mathrm{OW}} + k_4 \cdot R_{\mathrm{Water}} \quad (7)$$

where R_{Mix} represents the reflectance of the mixing pixel; R_{Oil}, R_{WO}, R_{OW}, and R_{Water} represent the reflectance of the simulated NEO, WO, OW, and seawater, respectively; k_1, k_2, k_3, and k_4 represent the horizontal area fraction of each target, with $k_1 + k_2 + k_3 + k_4 = 1$. In this article, the target category corresponding to the maximum horizontal area fraction k is taken as the category of mixing pixels. The partial oil-spill mixing spectral curves using Landsat-5 TM data as an example are shown in Fig. 6.

III. Model and Method

In this article, the cross-spatiotemporal remote sensing detection model coupled with the oil-spill optical and thermal response characteristics is constructed to investigate marine oil-spill cross-spatiotemporal detection in complex scenarios.

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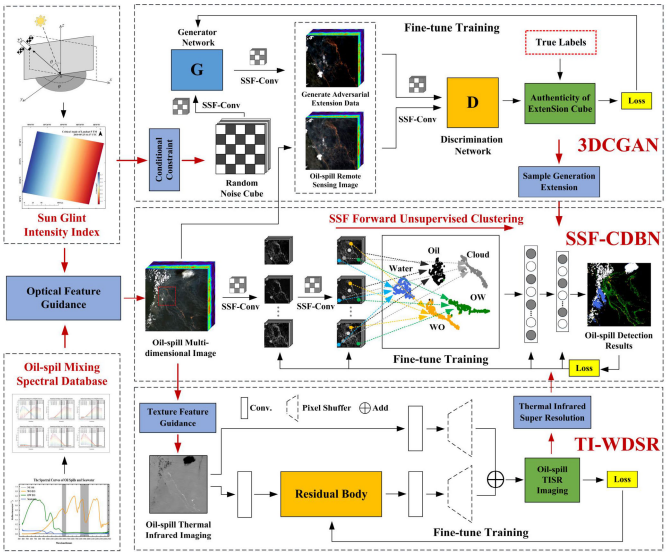


Fig. 7. Structure of the constructed oil-spill cross-spatiotemporal remote sensing detection model.

The model is constructed by adopting a modular idea, and it consists of four modules: OFG, SGII-constrained 3DCGAN, TI-WDSR, and SSF-CDBN.

The model takes SGII as a constraint to constrain 3DCGAN to generate oil-spill expansion samples under different sun glint intensities, thereby improving the diversity of the input samples of SSF-CDBN and model generalization. TI-WDSR is applied to enhance the super-resolution of low-resolution thermal infrared images, enhance the spatial distribution characteristics of oil-spill thermal infrared images, and improve the detection capabilities of the SSF-CDBN under the interference of complex clouds and fog. The optical extension samples generated by 3DCGAN and the thermal infrared samples after TI-WDSR super-resolution enhancement are used as the optical and thermal input data of the model, respectively. In this article, multiple oil-spill remote sensing images acquired in different times and spaces, different sun glint conditions, and different complex scenes are used to train the detection model. SGII is used to guide the model to detect oil spills in scenes with different sun glint intensities, and OMSD is used to guide the model to detect oil spills in mixing pixels. Under the joint guidance of the OFG composed of SGII and OMSD, SSF clustering and integrated extraction of different oil-spill targets are realized based on SSF-CDBN, thereby achieving marine oil-spill cross-spatiotemporal migration detection. The structure diagram of the constructed model is shown in Fig. 7.

A. Feature Extraction Module Based on SSF-CDBN

CDBN, which has the advantages of both unsupervised learning and supervised learning, is widely used in various deep learning tasks. The traditional CDBN extracts feature based on 2-D planar convolution kernels, but remote sensing data are mostly 3-D spectral cube data. Relying on 2-D convolution kernels will result in the separation of spatial and spectral features [43], [44], [45]. To address the issues of poor SSF feature extraction ability and weak generalization of traditional algorithms in complex real oil-spill scenarios,

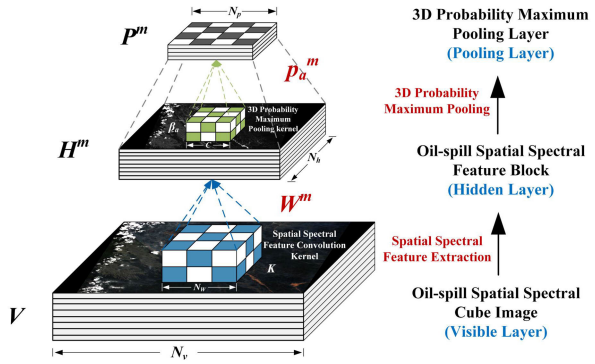


Fig. 8. Feature extraction process of 3DCRBM.

SSF-CDBN is proposed, which is composed of a 3-D convolution restricted Boltzmann machine (3DCRBM) proposed for the first time in this article.

In view of the combination of oil-spill SSF of remote sensing images of satellites, the module uses the SSF convolution kernel with a dimension of $N_w \times N_w \times N_w$ (K) and strengthens the clustering and integrated extraction ability of oil-spill high-level SSF information in complex scenarios by using the mechanism of feature forward unsupervised clustering and reverse fine-tuning. The feature extraction process is illustrated in Fig. 8.

The input layer (V) of the 3DCRBM is the oil-spill SS cube image of size $N_v \times N_v \times N_v$. The hidden layer (H) is composed of M SSF blocks of size $N_h \times N_h \times N_h$, and the energy function is shown in the following equation:

$$E(v,h) = -\sum_{m=1}^{M}\sum_{i,j,k=1}^{N_h}\sum_{r,s,t=1}^{N_w} h_{i,j,k}^{m} W_{r,s,t}^{m} v_{i+r-1,j+s-1,k+t-1} - \sum_{m=1}^{M} b_m \sum_{i,j,k=1}^{N_h} h_{i,j,k}^{m} - c\sum_{i,j,k=1}^{N_v} v_{i,j,k} \quad (8)$$

where $N_w \triangleq N_v - N_h + 1$, with v being the vector of V, h being the vector of H; $v_{i,j,k}$ represents the value of the nodes at position (i, j, k) of V, $v_{i+r-1,j+s-1,k+t-1}$ represents the value of the node after V slides (r, s, t) in the SS dimension, $h_{i,j,k}^{m}$ represents the value of the node at (i, j, k) of the mth SSF block of H, $W_{r,s,t}^{m}$ represents the connection weight between V and the mth SSF block, c represents the shared bias of V, and b_m represents the shared bias of the mth K.

Since the input information of V in the module is the oil-spill remote sensing reflectance SSF cube, the energy function is reconstructed to extract SSF from real-valued data. The simplified formula is shown in the following equation:

$$E(v,h) = \frac{1}{2}\sum_{i,j,k=1}^{N_v} v_{i,j,k}^{2} - \sum_{m=1}^{M} h^{m} \cdot \left(\tilde{W}^{m} * v\right) - \sum_{m=1}^{M} b_m \sum_{i,j,k=1}^{N_h} h_{i,j,k}^{m} - c\sum_{i,j,k=1}^{N_v} v_{i,j,k} \quad (9)$$

where h_m denotes the mth SSF block of H, $\tilde{W}^{m}$ is the weight of the mth feature block, and $*$ denotes the convolution operation. The joint probability distribution of (v, h) can be

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calculated based on the energy function, as shown in the following equation:

$$P(v,h)=\frac{e^{-E(v,h)}}{\sum_{(v,h)}e^{-E(v,h)}}. \tag{10}$$

When the node state h of H is determined, the activation probability of the node of V is shown in (11). Here, σ is the sigmoid function, $\sigma(x)=1/(1+e^{x})$

$$P\left(v_{i,j,k}=1\middle|h\right)=\sigma\left(\sum_{m}^{M}\left(W^{m}*h^{k}\right)_{i,j,k}+c\right). \tag{11}$$

When the node state v of V is determined, the activation probability of the node $h^{m}_{i,j,k}$ of H is shown in the following equation:

$$P\left(h^{m}_{i,j,k}=1\middle|v\right)=\sigma\left(\left(\tilde{W}^{m}*v\right)_{i,j,k}+b_{m}\right). \tag{12}$$

3DCRBM can construct the mapping relationship of the model by sampling the shared parameters of all hidden layers. The sampling method is the contrastive divergence algorithm, which can improve the efficiency of model feature extraction. The 3-D maximum pooling layer commonly used in traditional convolutional neural networks (CNNs) cannot achieve probabilistic reasoning and obtain the posterior probability required by the model proposed in this article. Therefore, a 3-D probability maximum pooling layer (3DPMP) is constructed to extract high-level SSF information about oil spills

$$\beta_{\alpha}\left\{(i,j,k):h^{m}_{i,j,k}\text{ belongs to the block }\alpha\right\}. \tag{13}$$

The SSF block of H consists of 3DPMP kernels (β) of size $\alpha\ C\times C\times C$. Assuming that the size of 3DPMP is $N_p\times N_p\times N_p$, where $C=N_H/N_P$, the constraints of 3DPMP are shown in (13). If and only if the SSF block in H is activated, the module can activate 3DPMP upward, and at most one unique value in the β_{α} block is activated into 3DPMP and mapped to a binary unit cube

$$\begin{aligned}E(v,h)=&\frac{1}{2}\sum_{i,j,k=1}^{N_v}v^{2}_{i,j,k}-\sum_{m=1}^{M}h^{m}\cdot\left(\tilde{W}*v\right)\\&-\sum_{m=1}^{M}b_{m}\sum_{i,j,k=1}^{N_h}h^{m}_{i,j,k}-c\sum_{i,j,k=1}^{N_v}v_{i,j,k}\\&\left(\text{subject to}\sum_{(i,j,k)\in\beta_{\alpha}}^{N_h}h^{m}_{i,j,k}\le 1\quad\forall m,\alpha\right).\end{aligned} \tag{14}$$

3DPMP can increase the module's perception range of marine oil-spill area, mitigate the influence of discrete information in V on high-level feature extraction, control module parameter elaboration, reduce parameter redundancy, and improve module detection speed and generalization. The 3DCRBM energy function equipped with 3DPMP is shown as (14)

$$P\left(p^{m}_{\alpha}=1\middle|v\right)=\frac{e^{I\left(h^{m}_{i,j,k}\right)}}{1+\sum_{(i',j',k')\in\beta_{\alpha}}^{N_h}e^{I\left(h^{m}_{i,j,k}\right)}}$$

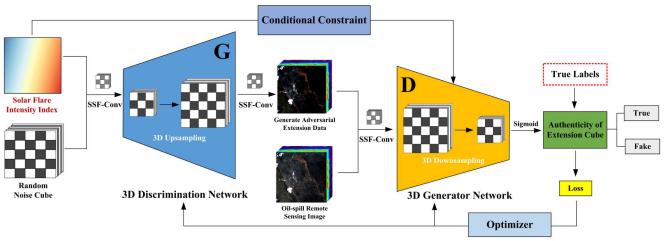


Fig. 9. Structure of 3DCGAN module.

$$I\left(h^{m}_{i,j,k}\right)\triangleq\sum_{m}^{M}\left(\tilde{W}^{m}*h^{m}\right)_{i,j,k}+b_{m}. \tag{15}$$

Based on the 3DCRBM energy function after adding probability constraints, the probability of the top 3DPMP node activated by the Softmax function is shown in (15), where $-I(h^{m}_{i,j,k})$ denotes the energy increment caused by the activated node $h^{m}_{i,j,k}$, and p^{m}_{α} represents the binary unit in the 3DPMP

$$\text{Loss}=\text{Maximize}_{(W,b,c)}\left[\mathcal{L}_{\text{likelihood}}(W,b,c)+\mathcal{L}_{\text{sparsity}}(W,b,c)\right]. \tag{16}$$

The module convergence direction is designed to maximize the log-likelihood value ($\mathcal{L}_{\text{likelihood}}$) of energy, and based on this, the penalty phase ($\mathcal{L}_{\text{sparsity}}$) of $\mathcal{L}_{\text{likelihood}}$ is added to limit the average activation characteristics of units of H, reduce the module parameters through sparse regularization, and enhance the efficiency and robustness of the feature extraction process, as shown in (16).

B. Oil-Spill 3-D SSF Cube Generative Adversarial Extension Module Constrained by SGII

The oil-spill optical remote sensing data under different sun glint intensities vary greatly. However, the amount of pure sample data in real oil-spill accidents is insufficient, and labeling training samples is time-consuming and the diversity of the sample is usually poor. To solve the above problems in remote sensing oil-spill detection at the present stage, the 3DCGAN constrained by SGII is constructed in this article to expand the SSF cube data of oil spills and seawater under different sun glint intensities and scenarios, increase the diversity of samples, and thus improve the generalization and robustness of the oil-spill detection model.

The 3DCGAN is composed of a generator (G) and a discriminator (D). The goal of D is to judge the authenticity probability of the input sample, and based on the probability, guide G to learn the distribution of the real oil-spill sample and generate an extended sample similar to the real sample. Different from the traditional GAN and DCGAN networks, the 3DCGAN proposed in this article is based on the 3-D SSF convolution kernel, which can fully integrate and extract oil-spill SSF information, maximize the probability of D, identify sample sources through adversarial training, and enable G to generate extended samples with maximum similarity to the real oil-spill SSF cube in spectral and spatial dimensions. The module structure is presented in Fig. 9.

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The SGII is utilized to characterize the sun glint intensity in optical satellite remote sensing images, which is an important indicator that changes the contrast between the reflectance of oil spills and background seawater. In this article, SGII is introduced into 3DCGAN as the sample expansion constraint to constrain the module to expand the oil and seawater SSF blocks and stabilize the convergence direction of the module. The training process of 3DCGAN is shown in the following equation:

$$\begin{aligned}\mathrm{Min_G Max_D}\ \{V(\mathrm{D},\mathrm{G}) = E_{x\sim p_{\mathrm{data}}(x)}\left[\log\left(\mathrm{D}\left(x\middle|\mathrm{SGII}\right)\right)\right] \\ + E_{z\sim p_z(z)}\left[\log\left(1-\mathrm{D}\left(\mathrm{G}\left(z\middle|\mathrm{SGII}\right)\right)\right)\right]\}\end{aligned} \quad (17)$$

where x represents the real oil-spill SSF cube, p_{data} represents the real data distribution, and $E_{x\sim p_{\mathrm{data}}(x)}$ represents the distribution that x is expected to obey. z represents a random 3-D Gaussian noise cube, p_z represents the generated data distribution, and $E_{z\sim p_z(z)}$ represents the distribution that z is expected to obey.

During the training process to optimize the D net, when the input of D is a real oil-spill sample x, the aim of 3DCGAN is to modify $\mathrm{D}(x)$ so that it approaches 1. When the input sample of D is the generated sample $\mathrm{G}(z)$, 3DCGAN aims to modify $\mathrm{D}(\mathrm{G}(z))$ to approach 0, i.e., $1\text{-}\mathrm{D}(\mathrm{G}(z))$ tends to 1, so D is maximized. During the training process to optimize the G net, when the input information is a random noise cube z, the module aims to modify $\mathrm{D}(\mathrm{G}((z))$ to approach 1, i.e., $1\text{-}\mathrm{D}(\mathrm{G}(z))$ tends to 0, so G is minimized. If and only if $p_z = p_{\mathrm{data}}$, there is a globally optimal solution to the problem of maximizing the two-sided game, and in this case, the 3DCGAN reaches the Nash equilibrium point and outputs oil-spill SSF cube extended samples with the best quality [35].

C. Super-Resolution Enhancement Module for Oil-Spill Thermal Infrared Images Guided by Texture Feature

Constrained by the complex radiation transmission mechanism and sensor hardware limitations of thermal infrared remote sensing, thermal infrared remote sensing images generally have a low spatial resolution. The TI-WDSR is proposed to improve the spatial resolution of BT data and oil-spill thermal infrared spatial geometric features, providing auxiliary support for oil-spill optical remote sensing cross-spatiotemporal detection under complex clouds and fog.

In this article, the WDSR is adopted as the basic algorithm for thermal infrared super-resolution, which has won the championship of the global super-resolution competition new trends in image restoration and enhancement (NTIRE). It has been regarded as one of the leading indicators on various datasets, and it is widely used in various super-resolution reconstruction tasks. Based on the team's previous research, it is found that the entropy feature in oil-spill texture features has the best oil-spill geometric characteristics and oil-water separability [23]. By using the gray level co-occurrence matrix (GLCM), the entropy texture features in the four directions of the optical remote sensing image are calculated and averaged, thereby guiding TI-WDSR to increase the spatial resolution of the thermal infrared data to 30 m, which is the same as that of the optical image.

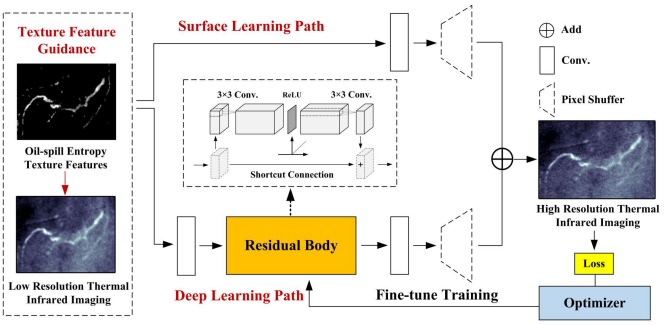


Fig. 10. Structure of TI-WDSR module.

TI-WDSR consists of two learning paths: the surface learning path and the deep learning path, and the module structure is shown in Fig. 10. The surface learning path extracts surface rough features of thermal infrared images through convolutional layers and subpixel convolutional layers. The deep learning path is based on stacked multilayer residual networks, and it has strong deep feature mining capabilities while suppressing the issues of gradient disappearance and gradient explosion. TI-WDSR can realize residual learning of the oil-spill deep geometric feature details under the guidance of the texture feature information. After upsampling through the subpixel convolution layer, the deep and surface features are accumulated and integrated to generate high-resolution thermal infrared images, as shown in the following equation:

$$I^{\mathrm{SR}} = \mathrm{UP}\left(I^{\mathrm{LR}} * W_{\mathrm{S}}\right) + \mathrm{UP}\left\{\left[\sum_{i=0}^{N}\left(F_i * W_i\right)\right] * W_{\mathrm{D}}\right\} \quad (18)$$

where I^{SR} represents the high-resolution oil-spill thermal infrared image after super-resolution, I^{LR} represents the original low-resolution thermal infrared image, W_{S} represents the weight of the surface learning convolution layer, W_{D} represents the weight of the deep learning convolution layer, $(F_i * W_i)$ represents the deep feature convolution extraction process of the ith residual module, and UP represents subpixel convolution upsampling.

The nonlinear rectified linear unit (ReLU) layer prevents the transmission of feature information from the surface layer to the deep network. The residual module of TI-WDSR takes linear low-rank convolution layers as the basic building blocks, and it increases the number of feature maps in front of the activation layer to improve feature transfer efficiency. In this way, it makes wider activation possible and enables the low-level low-resolution features from the surface layers to be effectively transferred to deep networks while ensuring the high nonlinearity of the deep network. The processing method of step-by-step channel downgrading is adopted by the residual block of TI-WDSR to reduce computational redundancy, and it is superimposed and fused with the original features to form a complete residual block to enhance the super-resolution efficiency and quality of thermal infrared images.

TI-WDSR performs image upsampling based on subpixel convolution, as shown in Fig. 11. Assuming that the target image is enlarged r times, the module will generate r^2 feature maps and reorganize the pixels at the same position on the r^2 feature maps adjacently in a certain order to generate a

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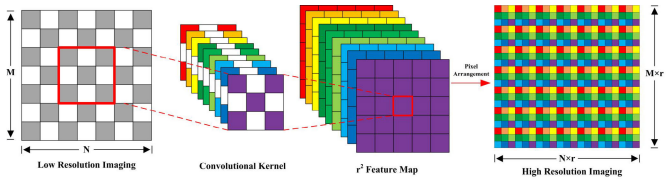


Fig. 11. Process of upsampling of TI-WDSR.

high-resolution image, thereby decreasing the calculation amount of deconvolution and interpolation and reducing human factor interference.

The weight normalization (WN) method is utilized by TI-WDSR to improve the module convergence speed. WN can decouple the vector length and vector direction of the weight, re-parameterize the weight coefficient, decrease the module calculation amount, improve the module response speed and super-resolution efficiency, and the training process will not produce additional dependencies. Compared with the traditional batch normalization (BN) method, WN can avoid the loss of information such as image brightness and contrast, and it improves the image super-resolution reconstruction effect. Assuming that the output is y, the process of WN is shown in the following equation:

$$y = w * x + b,\ w_n = \frac{g}{\|v\|}v \tag{19}$$

where w represents a k-dimensional weight vector, x represents a k-dimensional vector of input features, and b is a scalar bias term. w_n represents the reparameterized weight vector, v represents a k-dimensional weight vector, $\|v\|$ represents the Euclidean norm of v, and $\|w\| = g$, which is independent of v.

D. Accuracy Evaluation Index

Overall accuracy (OA) and Kappa coefficient are chosen in this study to evaluate the overall oil-spill detection accuracy and the consistency of the result. F_1-score considers both recall and precision, and it reduces the imbalance between the two. Therefore, F_1-score is selected to evaluate the detection accuracy of each single oil-spill target, and the calculation formula is shown in (20)–(24)

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \times 100\% \tag{20}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \times 100\% \tag{21}$$

$$F_1\text{-score} = 2 \cdot \frac{\text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}} \tag{22}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TN}+\text{TP}+\text{FN}+\text{FP}} \times 100\% \tag{23}$$

$$\text{Kappa} = \frac{\text{TP}+\text{TN}}{(\text{TN}+\text{FP}) \cdot (\text{TN}+\text{FN}) \cdot (\text{FN}+\text{TP}) \cdot (\text{FP}+\text{TP})} \tag{24}$$

where TP represents the number of true positives, TN represents the number of true negatives, FP represents the number of false positives, and FN represents the number of false negatives.

In addition, the Silhouette coefficient (SC), Davies–Bouldin index (DBI), and Calinski–Harabaz (CH) were selected as evaluation indicators for the feature extraction effect of the oil-spill SSF extraction module [46], [47], [48]. Spectral angle mapping (SAM), peak signal-to-noise ratio (PNSR), and structural similarity index measurement (SSIM) are adopted to evaluate the results of oil-spill SSF generative adversarial expansion from the spectral and spatial dimensions of the expansion sample [36], [49].

IV. Results

A. Selection of Basic Model for Cross-Spatiotemporal Feature Extraction of Marine Oil Spills

In this article, the DWH oil-spill accident in the Gulf of Mexico of the U.S. was selected as an example application scenario from the multiscene model training scenario. The scenario acquired by the Landsat-5 TM sensor on May 25, 2010, was complex and included NEO, WO, OW, seawater, clouds, and other typical oil-spill targets. It has been widely used and investigated by internationally renowned scholars [15], [38], [42], which is representative of data and model application.

Based on the Tensorflow framework, this study built deep belief network (DBN), deep neural network (DNN), CNN, CDBN, and other typical feature extraction deep learning models to perform oil-spill remote sensing detection in the research area, and compared and analyzed the results of the SSF-CDBN proposed in this article, thereby selecting the basic oil-spill feature extraction model. The experiment was conducted on the Dell-Precision 7820Tower workstation equipped with an Xeon-Gold 5122 processor and an NVIDIA-RTX2080Ti graphics card. The sample number of seawater, NEO, WO, OW, and clouds was 200 randomly selected, with a total of 1000. The experimental accuracy was calculated by taking the average value of three experimental results, as listed in Table II.

As shown in Table II, the 1D-CNN model based on 1-D convolutional feature extraction and feature map weight sharing obtains the highest oil-pill detection efficiency but unsatisfactory detection results. By combining the mechanism of feature forward unsupervised clustering and reverse fine-tuning, the SSF-CDBN proposed in this article can enhance the clustering and integrate the extraction ability of high-level oil-spill SSF information in complex scenarios, with the OA value of 89.53% and the Kappa coefficient of 0.863, both of which are superior to those of the other six types of traditional feature extraction deep learning models. Also, it demonstrates higher detection capabilities of WO and OW than other models, with the F_1-score reaching 0.838 and 0.935, respectively. Limited by the complex feature extraction mechanism, the detection time of SSF-CDBN is 11.89 min. The detection time is increased by about one-tenth, compared with 3D-CNN, which is the mainstream feature extraction model for remote sensing images.

As shown in Fig. 12, DNN, DBN, and 1D-CNN models detect marine oil-spills based on pixel-by-pixel spectral feature extraction methods but ignore the oil-spill spatial distribution characteristics, resulting in unsatisfactory detection results and obvious discrete "spots." 2D-CNN and CDBN consider

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TABLE II
ACCURACY OF THE SELECTION EXPERIMENT OF THE OIL-SPILL FEATURE EXTRACTION BASIC MODEL

Index	Feature Extraction Basic Model						
	1D-CNN	2D-CNN	3D-CNN	DNN	DBN	CDBN	SSF-CDBN
1#-Water	0.856	0.845	0.867	0.793	0.816	0.874	**0.901**
2#-NEO	0.835	0.831	**0.861**	0.804	0.850	0.820	0.851
3#-WO	0.552	0.575	0.686	0.322	0.464	0.815	**0.838**
4#-OW	0.832	0.886	0.922	0.829	0.828	0.910	**0.935**
5#-Cloud	0.770	0.859	0.873	0.803	0.837	0.862	**0.873**
OA/%	81.35	83.64	87.04	78.59	80.51	86.69	**89.53**
Kappa	0.755	0.786	0.830	0.716	0.744	0.827	**0.863**
Time/min	**6.57**	7.89	10.26	7.85	9.48	10.59	11.89

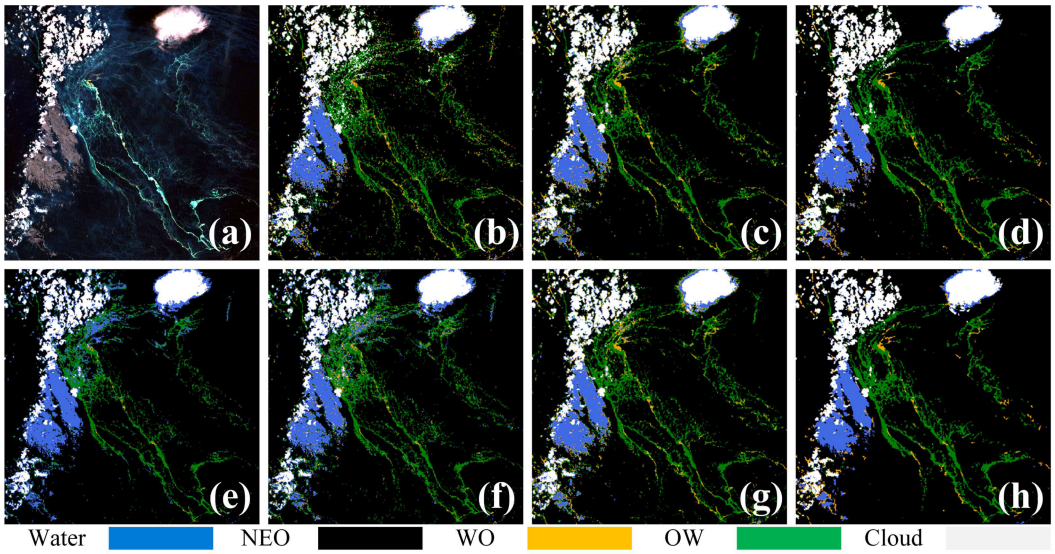


Fig. 12. Results of the selection experiment of the oil-spill feature extraction basic model. (a) Oil-spill remote sensing image of the research area. (b)–(h) Detection results of 1D-CNN, 2D-CNN, 3D-CNN, DNN, DBN, CDBN, and SSF-CDBN, respectively.

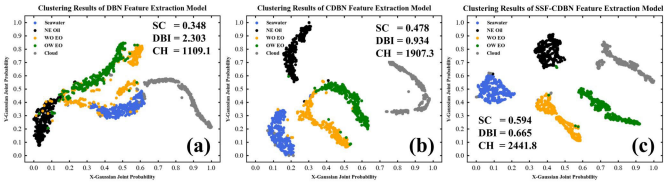


Fig. 13. Clustering results of the feature extraction model. (a)–(c) Extraction results of DBN, CDBN, and SSF-CDBN, respectively.

the spatial geometric features of oil-spill targets in feature extraction, and the uniformity of detection results is greatly enhanced, but there are still obvious misclassification phenomena in the detection around clouds. Compared with the 3D-CNN model, the SSF-CDBN proposed in this article can improve the integrated extraction ability of high-level oil-spill SSF in complex scenarios based on 3DCRBM, and it increases the sensing range of oil-spill areas by using 3DPMP, mitigates the influence of discrete information changes on high-level feature extraction, and substantially improve the oil-spill detection results from a visual perspective.

In this article, the feature forward unsupervised extraction effects of DBN, CDBN, and SSF-CDBN models are analyzed. It can be seen from Fig. 13 that owing to the 3-D SSF convolution integrated extraction mechanism, the oil-spill features extracted by SSF-CDBN have the best clustering effect and the strongest subjective visual interclass separability. According to the objective results, the SC and CH of SSF-CDBN are higher than those of DBN and CDBN, and the DBI value is only 0.665, showing an excellent SSF clustering ability. In terms of the model's oil-spill SSF extraction and detection capabilities, this article believes that SSF-CDBN has the best potential for marine oil-spill cross-spatiotemporal detection.

B. Oil-Spill 3-D SSF Cube Generative Adversarial Extension Constrained by SGII

The 3DCGAN constrained by SGII is proposed to increase the number of oil-spill SSF samples, enhance sample diversity, and improve the generalization and robustness of the model in different scenarios and sun glints. To verify its effectiveness, this article built traditional generative adversarial expansion models GAN and DCGAN, conducted generative adversarial expansion on real oil-spill remote sensing data, and compared and analyzed the results with the 3DCGAN.

Compared with the 1-D and 2-D oil-spill samples extended by GAN and DCGAN, the 3DCGAN proposed in this article can generate the 3-D SSF cube as the direct input data of SSF-CDBN. The generated SSF samples in the spatial dimension are averaged to obtain the spectral curve of the central pixel, and its expansion effect in the spectral dimension is analyzed. As shown in Fig. 14, from a visual perspective,

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TABLE III
QUALITY EVALUATION OF THE SAMPLE GENERATION EXTENSION EXPERIMENT

Index	SAM/°				SSIM				PNSR/dB			
Model	W	NEO	WO	OW	W	NEO	WO	OW	W	NEO	WO	OW
GAN	3.57	1.85	7.83	4.85								
DCGAN	2.05	1.89	4.72	3.24	0.96	0.98	0.64	0.55	36.27	44.87	24.72	18.24
3DCGAN	**1.43**	**1.54**	**1.40**	**1.31**	**0.99**	**0.99**	**0.92**	**0.94**	**47.37**	**48.89**	**36.53**	**35.33**

TABLE IV
ACCURACY OF THE SAMPLE GENERATION EXTENSION EXPERIMENT

Index	Number of Sample Generation Adversarial Extension					
	0	1000 (x1)	2000 (x2)	3000 (x3)	4000 (x4)	5000 (x5)
1#-Water	0.901	0.903	0.906	**0.921**	0.902	0.907
2#-NEO	0.850	0.865	0.886	**0.917**	0.892	0.852
3#-WO	0.838	0.922	0.941	**0.945**	0.940	0.863
4#-OW	0.935	0.944	0.959	**0.966**	0.963	0.941
5#-Cloud	0.873	0.898	0.891	**0.895**	0.893	0.881
OA/%	89.53	90.79	91.78	**93.15**	91.94	89.91
Kappa	0.863	0.880	0.892	**0.910**	0.895	0.868
Time/min	**11.89**	13.45	15.12	17.07	19.31	21.03

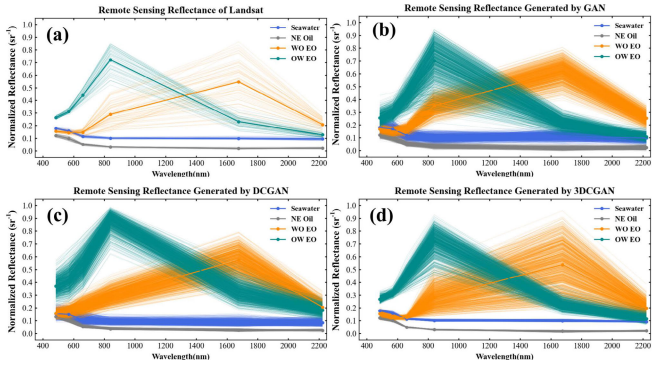


Fig. 14. Effects of the sample generation extension experiment in the spectral dimension. (a) Real spectral curves. (b)–(d) Results of GAN, DCGAN, and 3DCGAN, respectively.

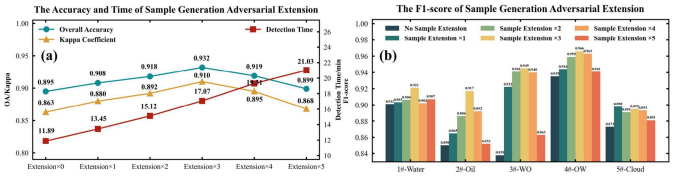


Fig. 15. Results of the sample generation extension experiment. (a) Accuracy and time of the sample extension experiment. (b) F_1-score of the extension experiment.

under the constraint of SGII, the results of 3DCGAN are more consistent with the distribution of real oil-spill spectra, and they are superior to those of the other two types of traditional generative adversarial expansion models in terms of sample diversity and richness.

Benefiting from the feature extraction method of 3-D SSF convolution, the 3DCGAN can integrate and mine the SSF information of real oil-spill samples and enhance the SSF combination and quality of the extended sample. As shown in Table III, from the perspective of the spectral dimension of the sample, the average SAM between the real and extended samples of the four types of oil-spill detection targets is only 1.42°, and the spectral similarity is extremely high. In terms of the spatial dimension, the average SSIM and PNSR of the extended sample reach 0.96 and 42.03 dB, respectively, demonstrating good sample quality. To sum up, the extended samples of 3DCGAN are better than those of the other two models in terms of subjective visual effects and objective accuracy, showing excellent oil-spill sample expansion capability.

Then, this article conducted experiments on the expansion number of oil-spill SSF samples, analyzed the variation trends of detection model performance, and determined the optimal sample expansion number. The number of training samples of the model in this training area is 1000. Based on the results of the pre-experiment, the real oil-spill samples were expanded by 1–5 times (the number of expansions is 1000–5000). As shown in Table IV, when the expansion factor is 3 (i.e., the number of expanded samples is 3000), the model shows the strongest oil-spill detection ability and result consistency, with the OA value reaching 93.15% and the Kappa coefficient reaching 0.910, and the detection accuracy F_1-score of single oil-spill target exceeding 0.917.

As shown in Fig. 15, as the number of oil-spill extended samples increases, the detection capability of the model is improved, both the OA value and the Kappa coefficient are improved to varying degrees, and the detection time also increases. When the number of samples reaches 3000, the oil-spill detection performance of the model reaches the peak. As the number of extended samples continues to increase, the model is overfitting, and the detection accuracy decreases. The F_1-score of the single oil-spill target has a similar variation trend to the overall detection accuracy, when the expansion factor is 3, the model achieves the best detection accuracy for each oil-spill target.

Therefore, the expansion factor of 3DCGAN is set to 3 to expand the oil-spill SSF cube. Also, it is believed that the detection ability of the model cannot be improved continuously by relying only on deep learning methods, and it is necessary to guide the model to detect oil-spill targets in complex scenarios based on the response mechanism of remote sensing and the optical characteristics.

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TABLE V
ACCURACY OF MARINE OIL-SPILL DETECTION GUIDED BY OFG

Feature	Index							
	1#-Water	2#-Oil	3#-WO	4#-OW	5#-Cloud	OA/%	Kappa	Time/min
MSI	0.921	0.917	0.945	0.966	0.895	93.15	0.910	**17.07**
MSI+SGII	0.948	0.964	0.944	0.964	0.949	94.73	0.931	18.38
MSI+OMSD	0.932	0.932	0.965	0.970	0.931	94.36	0.928	18.56
MSI+OFG	**0.954**	**0.968**	**0.966**	**0.971**	**0.953**	**96.16**	**0.951**	19.51

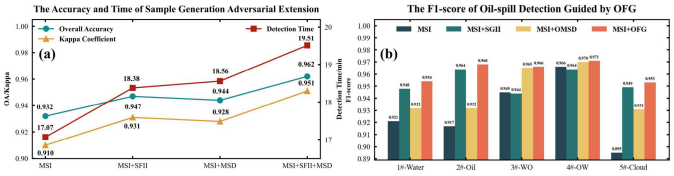


Fig. 16. Results of marine oil-spill detection guided by OFG. (a) Accuracy and time of oil-spill detection. (b) F_1-score of the experiment.

C. Remote Sensing Detection of Marine Oil Spills Guided by OFG

The sun glint intensity varies in different oil-spill accident scenarios, which results in different remote sensing reflectance of the same target, making oil-spill cross-spatiotemporal detection a bottleneck. Meanwhile, Landsat is a medium-scale resolution satellite to observe the oil-spill accident, and the mixing pixel problem is obvious. In this article, remote sensing observation geometry information is exploited to calculate the SGII, and the spectral linear mixture model is constructed to build the OMSD, thereby constructing the OFG to jointly guide the model to detect oil-spill mixing pixels under different scenarios, different observation geometry, and different sun glint intensities. The model application verification is conducted in typical research areas.

OFG-guided detection experiment is conducted based on the original multispectral (MSI) input data. As shown in Table V, under the guidance of single features SGII and OMSD, the model's overall oil-spill detection capabilities have been enhanced to varying degrees. Under the joint guidance of the OFG composed of SGII and OMSD, the detection performance reaches the peak, with the OA value reaching 96.16%, the Kappa coefficient reaching 0.951, and the F_1-score for the single oil-spill target reaching 0.966, demonstrating excellent marine oil-spill detection capabilities, but the model's calculation burden is further increased.

As shown in Fig. 16, under the guidance of SGII, the model can learn the positive and negative contrast rules of oil and seawater under different observation geometry and sun glint intensities, thereby greatly improving the detection accuracy of NEO and seawater. However, the detection accuracy of the two types of OE is almost unchanged, and this is because the change of the observed geometry factors has little effect on the remote sensing reflectance of emulsions. Under the guidance of OMSD, the detection accuracy of a single oil-spill target is enhanced to varying degrees, especially for OE, and this is because WO and OW are only distinguished by volume concentration differences, and the two are usually associated in real oil-spill accidents, so the problem of mixing pixels is more serious. Also, the model can accurately detect the mixing pixel category according to the proportion of each object in the pixel, which greatly enhances the detection accuracy of OE.

As illustrated in Fig. 17, compared with the oil-spill detection results guided by no OFG, the visual effect of the detection results under the guidance of OFG is greatly improved. The misclassification phenomenon at the boundary of oil and water, and the mixing pixel problem of OE are solved, and even the oil-spill detection ability under clouds and fog interference is further enhanced. Therefore, it can be concluded that under the guidance of OFG, the oil-spill detection model proposed in this article can accurately detect mixing pixels in complex scenarios, and it is expected to overcome the bottleneck of oil-spill cross-spatiotemporal detection under different scenarios and sun glint intensities.

D. Super-Resolution Enhancement of Oil-Spill Thermal Infrared Data Guided by Texture Feature

Aiming at the problem of low spatial resolution of thermal infrared satellite remote sensing data, TI-WDSR was built in this article to enhance the spatial resolution of thermal infrared BT images of oil spills and improve the model's ability to resist cloud and fog interference. Three types of typical super-resolution deep learning models super-resolution CNN (SRCNN), enhanced deep super-resolution network (EDSR), and WDSR were built, and the thermal infrared super-resolution results were compared and analyzed with those of the TI-WDSR proposed in this article.

Super-resolution enhancement comparative experiments were conducted in two experimental areas. It can be seen from Table VI that SRCNN establishes super-resolution mapping relationships based on full convolution feature extraction; it has a simple structure and high efficiency but obtains poor super-resolution results. Since the residual network idea was introduced into EDSR to enhance the super-resolution of the thermal infrared image, each precision evaluation index is improved to varying degrees. In this article, WDSR is selected as the basic thermal infrared super-resolution enhancement model. Under the guidance of the entropy texture feature, the super-resolution performance is further enhanced. The PNSR of the model in test area B reaches 42.58 dB, the SSIM reaches 0.99, and the root mean square error (RMSE) is only 0.007 K, demonstrating an excellent thermal infrared super-resolution enhancement effect.

As shown in Fig. 18, the thermal infrared data fitting effect of the two experimental areas is improved substantially, and the R^2 of TI-WDSR exceeds 0.99. As illustrated in the enlarged figures in Fig. 19, under the guidance of the entropy texture feature, the thermal infrared super-resolution

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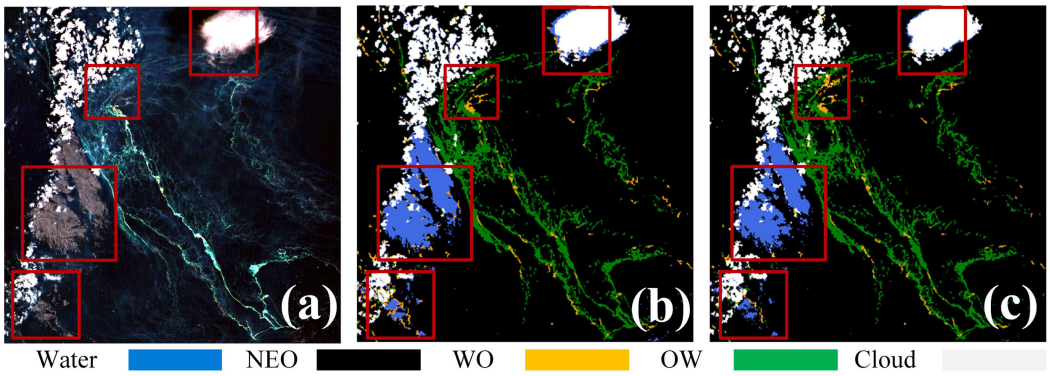


Fig. 17. Comparison results of marine oil-spill detection guided by OFG. (a) Oil-spill remote sensing image of the research area. (b) and (c) Detection results of the model on MSI and MSI + OFG, respectively.

TABLE VI
ACCURACY OF THE THERMAL INFRARED SUPER-RESOLUTION EXPERIMENT

Area	Test-A				Test-B			
Index	SRCNN	EDSR	WDSR	TI-WDSR	SRCNN	EDSR	WDSR	TI-WDSR
PNSR/dB	30.90	31.89	33.32	**38.88**	33.56	35.14	37.86	**42.58**
SSIM	0.914	0.926	0.947	**0.983**	0.946	0.957	0.977	**0.992**
R^2	0.965	0.972	0.979	**0.991**	0.955	0.968	0.983	**0.991**
RMSE/K	0.028	0.025	0.021	**0.011**	0.020	0.017	0.012	**0.007**
Time/min	**3.89**	9.54	8.76	9.40	**4.10**	9.05	8.89	9.30

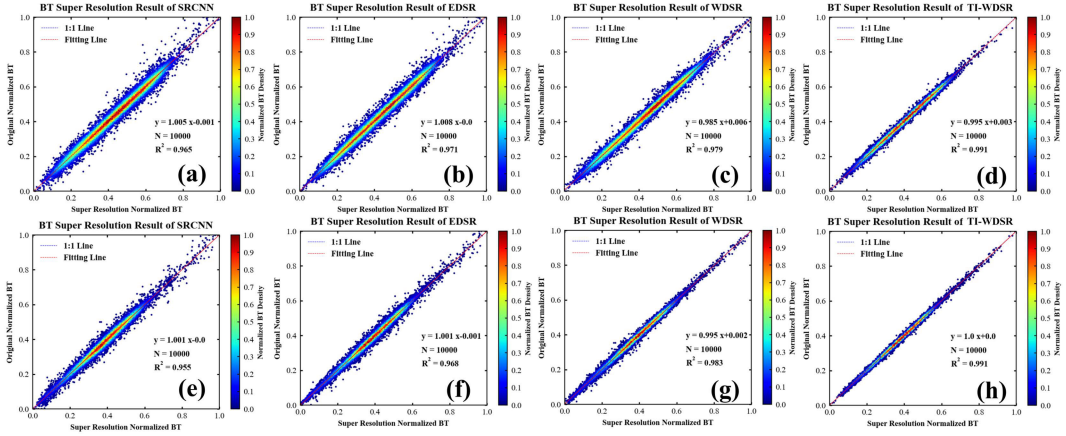


Fig. 18. Fitting effects of the thermal infrared super-resolution experiment. (a)–(d) Fitting effects of SRCNN, EDSR, WDSR, and TI-WDSR in Test-A, respectively. (e)–(h) Fitting effects in Test-B.

enhancement results of TI-WDSR are greatly improved at the visual level. The oil-spill thermal infrared image after TI-WDSR super-resolution has delicate spatial geometric characteristics similar to optical remote sensing images, the fuzzy distribution phenomenon at the oil-water boundary has been greatly suppressed, and the thermal infrared BT of oil spills has been effectively highlighted.

V. DISCUSSION

A. Contribution of Thermal Infrared Characteristic to Oil-Spill Detection Under Cloud and Fog Interference

When three-quarters of the marine optical remote sensing images are obscured by clouds and fog, the sea clouds and thin oil film under certain sun glint intensity and observation geometry all appear as bright targets in the optical images. Therefore, when conducting the cross-spatiotemporal detection of oil spills to verify the necessity of combining optical and thermal remote sensing technologies for oil spill detection, it was found that the detection model often led to a wrong separation of clouds and bright oil film. It can be seen from the migration detection accuracy in Table VII that the detection results based on single-dimensional optical data under cloud interference are not satisfactory, it fails to effectively identify cloud targets in the image, since both are highlighted targets in the optical data, and almost all of the targets are wrongly classified as oil spills. The OA value is only higher than 64.75%, the Kappa is greater than 0.517, and the F_1-score of single oil-spill targets is only higher than 0.667.

According to the optical and thermal response mechanism of oil spills described in Section II-C, clouds are low in

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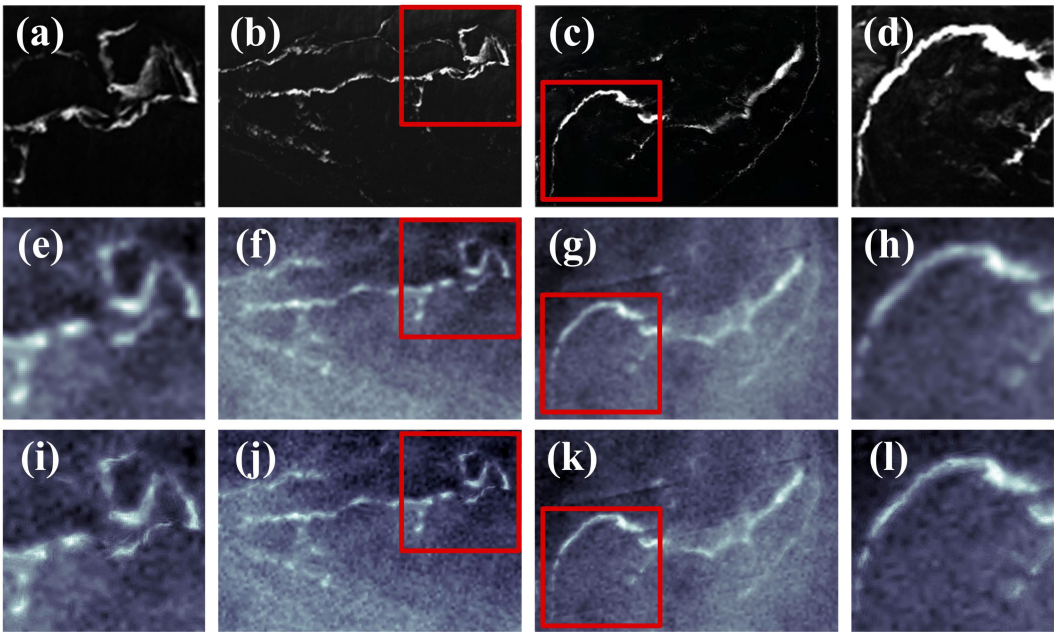


Fig. 19. Results of the thermal infrared BT super-resolution experiment. (b) and (c) NIR band of images of Test-A and Test-B, respectively. (f) and (g) Origin BT images of Test-A and Test-B. (j) and (k) Super-resolution enhancement results of TI-WDSR of Test-A and Test-B, respectively. Enlarged figures of (a), (e), and (i) Test-A and (d), (h), and (l) Test-B.

TABLE VII
ACCURACY COMPARISON OF THE MODEL BEFORE AND AFTER COUPLING OPTICAL AND THERMAL INFRARED FEATURES

Area	Test-A		Test-B	
Index	MSI	MSI+TI	MSI	MSI+TI
1#-Water	0.774	**0.818**	0.809	**0.855**
2#-NEO	0.676	**0.841**	0.667	**0.864**
3#-WO	0.766	**0.843**	0.757	**0.836**
4#-Cloud	0.00	**0.822**	0.00	**0.806**
OA/%	67.67	**83.05**	64.75	**84.37**
Kappa	0.544	**0.770**	0.517	**0.789**
Time/min	**6.57**	7.89	**6.26**	7.85

temperature and appear as dark targets in thermal infrared images. Therefore, in this article, the oil-spill optical and thermal response characteristics are coupled, and the thermal infrared data of oil spills after enhanced super-resolution and optical remote sensing data are jointly fed into the oil-spill detection model.

As shown in Table VII, after coupling the oil-spill optical and thermal response characteristics, although the detection time is slightly increased, the oil-spill detection capability is greatly improved. In the test areas, the OA value increases by 15.38% and 19.62%, the Kappa coefficients increase to 0.770 and 0.789, and the F_1-score of NEO increases by 0.226 and 0.272, respectively. As shown in Fig. 20, the oil-spill cross-spatiotemporal detection effect of the model has been enhanced substantially at the visual level based on SSF-CDBN proposed in this article, and the problem of misclassifying clouds into oil spills before coupling optical and thermal response characteristics is effectively solved.

B. Cross-Spatiotemporal Remote Sensing Detection of Marine Oil Spills Under Ideal Scenarios

In this article, optical and thermal remote sensing images of oil-spill accidents such as the 2010 U.S. Gulf of Mexico oil spill, the 2017 Persian Gulf oil spill, and the 2021 China Yellow Sea oil spill were obtained, leading to 12 typical oil-spill scenes in total. This article comprehensively considers the complexity of the oil-spill scene, the impact of sun glints, cloud interference, and other factors, and divides the oil-spill accident into two types of scenarios: ideal scenarios (Tests A–F) and complex scenarios (Tests G–L). Then, with the detection model proposed in this article, research on cross-spatiotemporal remote sensing detection of marine oil spills under different scenarios is conducted.

As illustrated in Fig. 21, due to the low complexity and weak cloud interference of the ideal oil-spill scenario, our proposed model can accurately detect the oil-spill distribution range and OE type. As shown in Fig. 21(i) of the oil-spill accident in the Yellow Sea of China in May 2021, the proposed model can accurately detect the distribution and pollution areas of NEO and WO, and it even detects towed oil traces formed when the clean-up ship passed through oil spills. However, the detection effect in the oil-spill scenario of DWH in May 2010 is not ideal, as shown in Fig. 21(k). Affected by factors such as oil film thickness, observation geometry, and sun glint intensity, the reflectance of oil film exhibits obvious light and dark differences, which poses challenges

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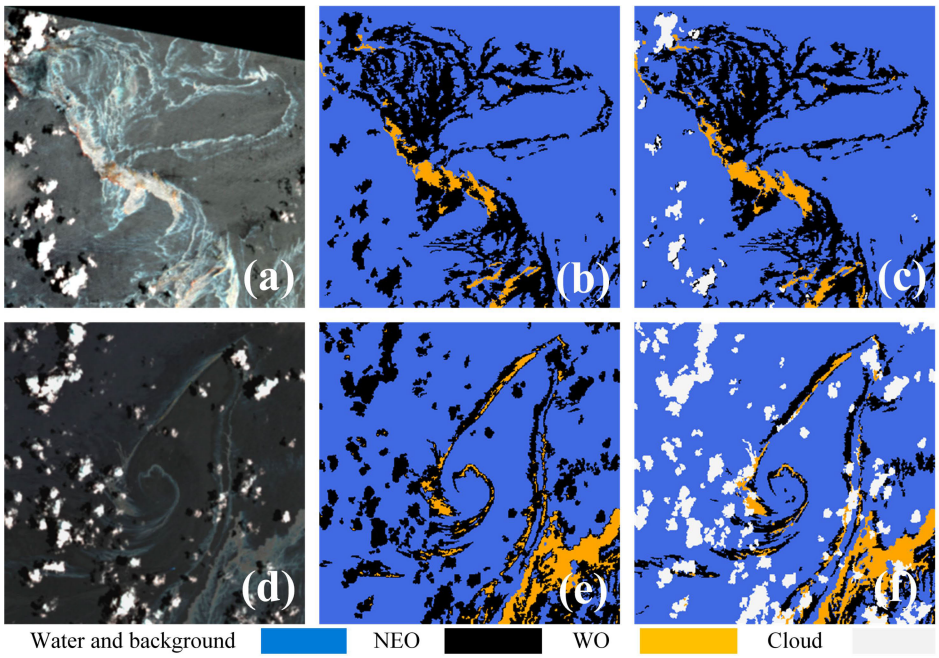


Fig. 20. Results comparison before and after coupling optical and thermal characteristics. (a) and (d) Remote sensing images of Test-A and Test-B, respectively. (b) and (e) Detection results of Test-A and Test-B based on single-dimensional optical data, respectively. (c) and (f) Detection results of Test-A and Test-B after coupling optical and thermal characteristics, respectively.

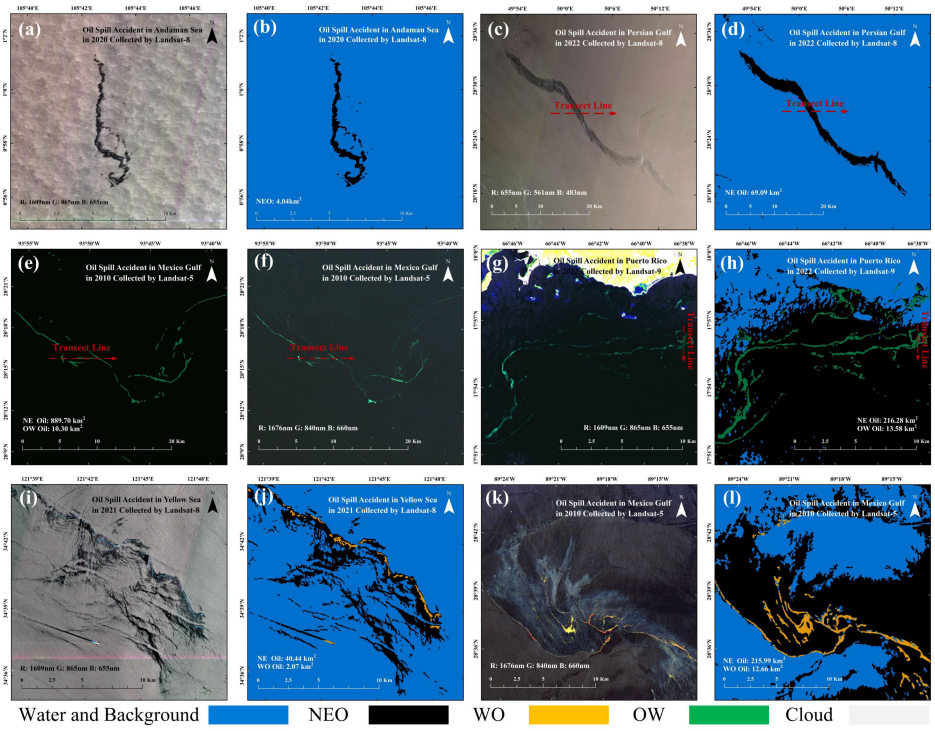


Fig. 21. Results of oil-spill cross-spatiotemporal detection in ideal scenarios (the land was masked into the background). (a), (c), (e), (g), (i), and (k) Oil-spill remote sensing images of Tests A–F, respectively. (b), (d), (f), (h), (j), and (l) Detection results of Tests A–F, respectively.

to the cross-spatiotemporal detection of the model. However, the model demonstrates stable robustness and accurately completes the oil-spill detection task in this scenario.

In this article, the accuracy of the model's cross-spatiotemporal detection results was evaluated. As listed in Table VIII, in the ideal oil-spill scenario, the proposed detection model performs well in the test areas A–F, with the highest OA value of the oil-spill target reaching 92.94%, the Kappa coefficient reaching 0.857, and the F_1-score reaching 0.921.

To verify and evaluate the cross-spatiotemporal detection results of the model more objectively, the detection results of three ideal oil-spill scenarios were randomly selected, and

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TABLE VIII
ACCURACY OF OIL-SPILL CROSS-SPATIOTEMPORAL DETECTION IN IDEAL SCENARIOS

No	Area	1#-Water	2#-NEO	3#-WO	4#-OW	5#-Cloud	OA/%	Kappa
1	Test-A	**0.936**	**0.921**				**92.94**	**0.857**
2	Test-B	0.925	0.910				91.83	0.835
3	Test-C		0.909		**0.913**		91.08	0.822
4	Test-D	0.901	0.880		0.873		88.78	0.816
5	Test-E	0.881	0.880	**0.844**			87.47	0.800
6	Test-F	0.852	0.859	0.802			84.46	0.758

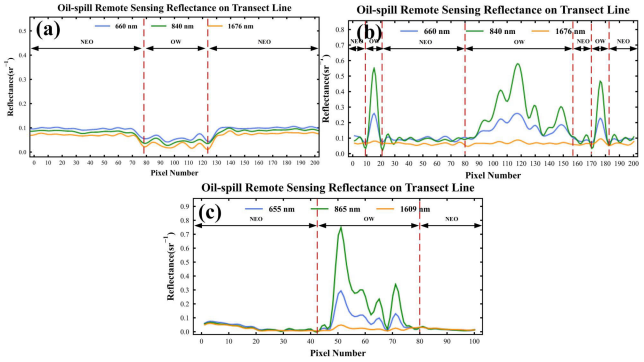


Fig. 22. Oil-spill transect line spectral response characteristics in ideal scenarios. (a), (b), and (c) Remote sensing reflectance on the transect line of Tests B, C, and D, respectively.

the spectral response characteristics of the transect lines were analyzed using red light, near-infrared, and short-wave infrared characteristic bands.

Fig. 22(a) shows the diagram of the oil-spill transect line spectral response characteristic of Test B. It can be seen from this figure that the remote sensing reflectance at the boundary between the oil spills and seawater changes obviously. Under the observation geometry and sun glint intensity corresponding to this scenario, the reflectance of seawater is greater than that of oil spills. Along the transect line direction are seawater, NEO, and seawater in order, which is consistent with the oil-spill distribution in real accidents. Also, in the spectral response characteristic diagram of Tests C and D in Fig. 22(b) and (c), along the transect line, there are obvious responses in the feature bands of near-infrared and red light, corresponding to the spectral response characteristics of OW. Along the transect line, NEO, WO, and NEO are in order.

C. Cross-Spatiotemporal Remote Sensing Detection of Marine Oil Spills Under Complex Scenarios

Based on the proposed cross-spatiotemporal remote sensing detection model coupling the oil-spill optical and thermal response characteristics, cross-spatiotemporal detection was conducted on the marine oil spills under complex scenarios, and the model's oil-spill detection capability under complex environments was tested.

As illustrated in Fig. 23(a), in the Persian Gulf oil spill accident in September 2021, the oil tanker caught fire and generated smoke due to the accident. Based on the optical and thermal response characteristics, our proposed model can identify and eliminate the smoke and detect the oil-spill target accurately under the interference of smoke. As shown in Fig. 23(c) and (e) of the DWH oil-spill accident in June 2010, after coupling optical and thermal response characteristics, our proposed model can accurately distinguish the clouds and oil film, which are also bright targets, to enhance the oil-spill detection accuracy. As illustrated in Fig. 23(k), the DWH oil-spill accident also occurred in June 2010. The oil-spill scenario was complex and seriously disturbed by clouds, and the oil-spill types included NEO, WO, and OW. In this scenario, our proposed model still shows stable detection capability; under the dual-drive of optical and thermal characteristics, the influence of clouds is suppressed, and the oil-spill area and OE type are accurately detected.

As listed in Table IX, in the complex oil-spill scenario, the proposed model also shows reliable detection ability, and the OA value of test areas G–L is higher than 80.11%, the Kappa coefficient is greater than 0.745, and the F_1-score exceeds 0.782. As shown in Fig. 24, compared with the ideal scenarios, due to the high complexity of the scenario and cloud interference, the model generally has a low detection accuracy in the complex scenario. In addition, the detection accuracy of the proposed model further decreases as the complexity of the scenario increases, e.g., the oil-spill scenario in Test L includes three types of oil-spill targets with seawater and cloud interference.

In this article, the cross-spatiotemporal detection results of three complex oil-spill scenarios were also randomly selected to analyze the spectral response characteristics of the transect line and verify the authenticity of the detection results.

As shown in the oil-spill transect line spectral response characteristic of Test G in Fig. 25(a), there are shortwave infrared and near-infrared response characteristics along the transect line, conforming to the spectral response law of WO, and the segmented spectral response characteristics of oil and water are obvious, so the detection results along the transect line are seawater, NEO, WO, NEO, and seawater in order.

It can be seen from the spectral response characteristic of Test-K in Fig. 25(b) that the response of short-wave infrared and near-infrared bands is obvious, corresponding to the response characteristics of WO. The response degree of the short-wave infrared band changes in the direction of the transect line, which is caused by the change of emulsion concentration and equivalent thickness of WO. Meanwhile, along the transect line of Test L, the short-wave infrared band of WO and the near-infrared band of OW respond in turn, and

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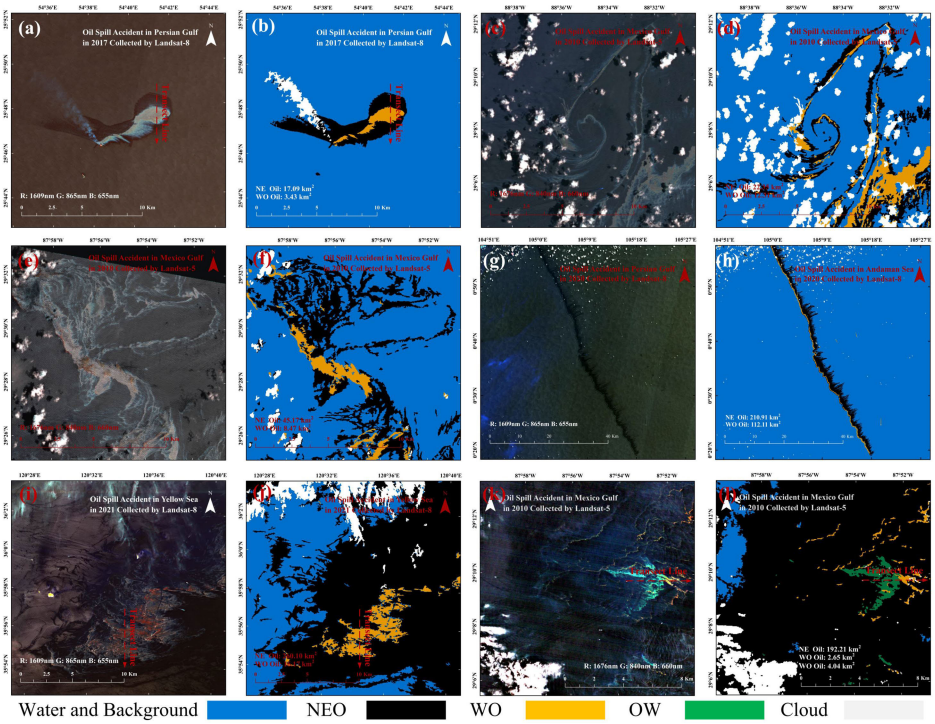


Fig. 23. Results of oil-spill cross-spatiotemporal detection in complex scenarios (the land was masked into the background). (a), (c), (e), (g), (i), and (k) Oil-spill remote sensing images of Tests G–L, respectively. (b), (d), (f), (h), (j), and (l) Detection results of Tests G–L, respectively.

TABLE IX
ACCURACY OF OIL-SPILL CROSS-SPATIOTEMPORAL DETECTION IN COMPLEX SCENARIOS

No	Area	1#-Water	2#-NEO	3#-WO	4#-OW	5#-Cloud	OA/%	Kappa
1	Test-G	**0.898**	0.808	0.816		0.792	**85.88**	0.766
2	Test-H	0.855	**0.864**	0.836		0.806	84.37	**0.789**
3	Test-I	0.818	0.841	**0.843**		**0.822**	83.05	0.770
4	Test-J	0.866	0.800	0.784		0.782	82.46	0.746
5	Test-K	0.824	0.803	0.793		0.787	80.33	0.748
6	Test-L	0.812	0.802	0.782	**0.795**	0.805	80.11	0.745

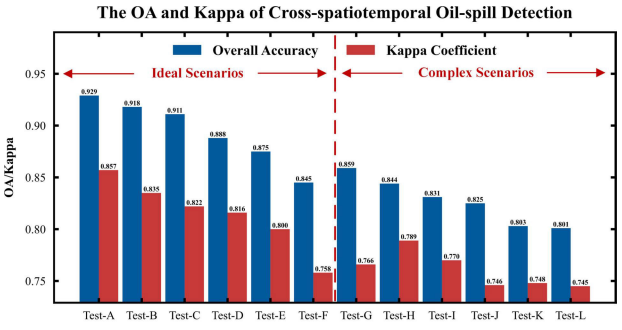


Fig. 24. OA and Kappa of cross-spatiotemporal oil-spill detection.

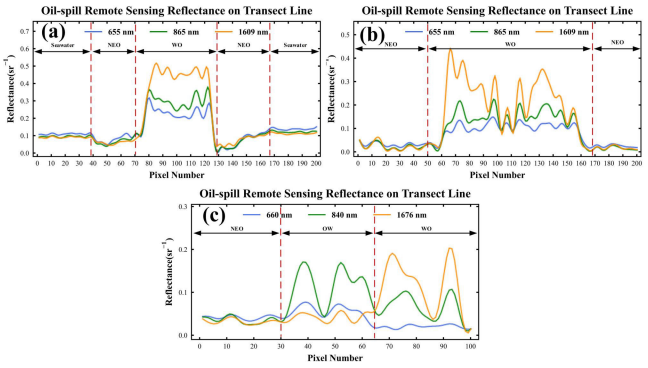


Fig. 25. Oil-spill transect line spectral response characteristics in complex scenarios. (a)–(c) Remote sensing reflectance on the transect line of Tests G, K, and L, respectively.

there are obvious spectral characteristic band changes at the boundary between the two, and the spectral characteristics in each distribution region conform to the response law of the two. Therefore, the detection results are NEO, OW, and WO in sequence, and they are consistent with the actual oil-spill scenario distribution.

The best-performing 3D-CNN and CDBN models in Section IV-A were selected to conduct oil-spill detection in complex scenes and compared with the SSF-CDBN model. As shown in Fig. 26, the OA and Kappa of SSF-CDBN in each test area are higher than those of the other two models. Benefiting from the mechanism of combined optical and thermal detection, the model has strong anti-cloud and fog interference ability, and its detection advantages in Test-G, Test-H, and Test-I are particularly obvious.

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TABLE X
COMPARISON OF DETECTION ACCURACY OF MODELS IN COMPLEX SCENES

No.	Model	3D-CNN		CDBN		SSF-CDBN	
	Area	OA/%	Kappa	OA/%	Kappa	OA/%	Kappa
1	Test-G	73.14	0.558	72.39	0.541	**85.88**	**0.766**
2	Test-H	62.57	0.487	63.19	0.495	**84.37**	**0.789**
3	Test-I	65.19	0.516	65.40	0.521	**83.05**	**0.770**
4	Test-J	78.32	0.729	76.57	0.710	**82.46**	**0.746**
5	Test-K	74.43	0.659	74.81	0.664	**80.33**	**0.748**
6	Test-L	70.38	0.618	69.43	0.607	**80.11**	**0.745**
AVE		70.67	0.595	70.30	0.590	**82.70**	**0.761**

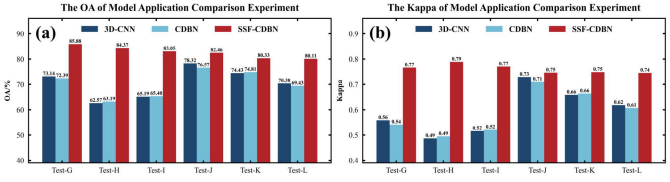


Fig. 26. Comparison of detection results of models in complex scenes. (a) OA of model application comparison experiment. (b) Kappa of model application comparison experiment.

As shown in Table X, the average OA and Kappa of the SSF-CDBN model in the six test areas are much higher than those of 3D-CNN and CDBN, reaching 82.70% and 0.761, respectively, which is a significant increase of about 12% and 0.17 compared with the two. Therefore, it is not difficult to conclude that the SSF-CDBN model proposed in this article has outstanding anti-cloud interference ability and obvious advantages in cross-spatiotemporal detection in complex oil-spill scenes.

The oil-spill detection model proposed in this article can generate the thematic map of oil-spill remote sensing detection based on the detection results. The thematic map contains key information such as the location of the accident, the time of the oil spill, the data source, and the type and area of oil-spill pollution. Therefore, the cross-spatiotemporal remote sensing detection model coupling the oil-spill optical and thermal response characteristics and the detection thematic map will provide new technical methods and key data support for the emergency treatment of marine oil-spill accidents.

D. Future Study

In the next stage of our study, we will use the 3-D mixing model proposed by Jiao to simulate the oil-spill mixing spectral in real oil-spill scenarios [41] supplement and enhance OMSD, and guide the detection model to detect oil-spill mixing pixels more accurately. In addition, we will apply satellite sensors of different types and scales to analyze the applicability of the proposed oil-spill detection model.

It is difficult to effectively distinguish algae from oil spills in MSI satellite images [16], and before conducting detection, most studies need to make comprehensive judgments based on expert interpretation experience, accident site data, marine pollution announcements, and other information. In the future study, the difference analysis of the optical and thermal response characteristics between algae and oil spills will be performed to enhance the detection model's ability to distinguish algae in MSI satellites.

At present, the complex cloud interference targeted by the model is low-level cumulus clouds. In the next stage, it will target more complex large-area stratus cloud interference to explore the applicability of the model method. In addition, the model is only for remote sensing detection of crude oil and its emulsions, and its applicability to other oil products is relatively weak. However, heavy fuel oil leakage is also one of the common cases of marine oil-spill accidents. In the next stage, it is planned to collect measured spectral data of fuel oil and its emulsions, enrich the oil-spill detection model, and carry out scientific verification and application.

In the future, based on the qualitative detection results of oil spills and emulsions, we will realize quantitative estimation of marine oil spills, inversion of oil film thickness and emulsion concentration, and obtain the oil-spill distribution and emulsion degree level at the accident site, thereby providing key information for pollution compensation accountability and on-site scientific decision-making.

VI. CONCLUSION

The frequency of oil-spill accidents is increasing year by year, oil spills and emulsions are highly detrimental to the ecological environment, economic development, and human health. In this article, a cross-spatiotemporal remote sensing detection model coupling the oil-spill optical and thermal response characteristics was proposed to improve the diversity of oil-spill samples and model generalization, enhance the thermal infrared BT resolution of oil spills and the ability to avoid cloud and fog interference, and realize the cross-spatiotemporal detection of oil-spill range and emulsion type. The conclusions of this article are provided as follows.

1) In this article, hyperspectral and thermal infrared observation experiments of oil spills were designed and conducted under controlled experimental conditions, the oil-spill optical and thermal response characteristics were analyzed, and the technical feasibility of oil-spill satellite remote sensing detection was validated. The OFG was constructed to guide the model to detect oil-spill mixing pixels under different sun glint intensities, and the OA and Kappa reached 96.16% and 0.951.
2) With SGII as the constraint, 3DCGAN was constrained to expand the oil-spill SSF samples under different sun glint intensities. The results indicated that the quality of

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the extended sample was reliable, the SAM was only 1.42°, and the average SSIM and PNSR were 0.96 and 42.03 dB, respectively. When the sample extension factor was 3, the model detection performance reached a peak, resulting in the OA and Kappa of 93.15% and 0.910, respectively.

3) The TI-WDSR was employed to achieve super-resolution enhancement of low-resolution thermal infrared images and improve the thermal infrared spatial geometric characteristics of oil spills, with R^2 reaching 0.99. The model's cross-spatiotemporal detection ability under complex cloud interference was improved, and the OA was increased by 15.38% and 19.62%, and Kappa was improved by 0.226 and 0.272, respectively.
4) The SSF-CDBN was built by coupling the optical and thermal response characteristics to achieve oil-spill cross-spatiotemporal remote sensing migration detection under different complex scenarios and generate the thematic map for oil-spill accident detection. The model demonstrated a stable generalization capability, with the OA and Kappa exceeding 84.46% and 0.758 in ideal scenarios, and 80.11% and 0.745 in complex scenarios.

To sum up, the cross-spatiotemporal remote sensing detection model coupling the oil-spill optical and thermal response characteristics proposed in this article can suppress the interference of clouds and other factors, overcome the bottleneck of oil-spill cross-spatiotemporal detection, and realize the accurate detection of oil spills and emulsions in complex scenarios. The proposed model is expected to make important contributions to the emergency treatment of marine oil-spill accidents.

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